

Management and Product Innovation

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1st Italian Macro with Micro Data Workshop

University of Naples Federico II, Naples

October 3, 2026

Management and research: AIM by Google

[Framework](#)[Evidence explorer](#)[Results](#)[Paper ↗](#)

01 / THE FRAMEWORK

A map of possibilities. A reason for the next experiment.

AIM searches over explicit research ideas. Inspired by Bayesian optimization, it separates understanding the idea space from choosing where to spend the experimental budget.

AGENTIC SURROGATE

What looks promising?

Organize. Group ideas by semantic research direction, even when they come from different generation lineages. Rebuild the map as the pool grows.

Estimate. Rank clusters and unevaluated ideas using observed scores, evidence gaps, novelty, and implementation lessons.

Ordinal ranks express relative promise—not calibrated reward predictions.

AGENTIC ACQUISITION

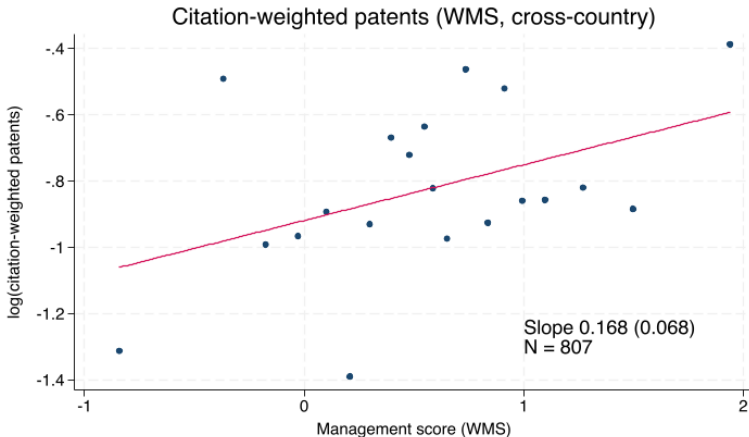
What should we try next?

Dispatch. Choose explore/exploit actions at both the cluster and idea levels, then select concrete ideas for parallel solvers.

Solve & Expand. Implement and evaluate the selected ideas. Use audited evidence to refine, combine, repair, or introduce new ideas.

A promising direction can still contain an unfamiliar idea worth exploring.

Management and patenting



Note: Authors' calculations based on World Management Survey manufacturing firms with at least one patent, merged to citation-weighted patents. The y-axis is log citation-weighted patents, residualized on industry and country fixed effects and on log employment, age and competition. Standard errors are robust.

Motivation

- ▶ Better management is associated with higher innovation.
- ▶ We know little about why.
- ▶ Three channels documented in the literature:
 - ▶ CEO ability: Acemoglu, Akcigit & Celik (2022); Custódio, Ferreira & Matos (2019).
 - ▶ Management as a technology: Bloom, Hartley, Sadun, Schuh & Van Reenen (2025).
 - ▶ Organization of R&D and delegation: Akcigit, Alp & Peters (2021); Acemoglu, Aghion, Lelarge, Van Reenen & Zilibotti (2007).
- ▶ This paper: the relationship between management practices and the innovation process.

This paper

- ▶ **Question:** how do management practices affect innovation? We focus on product innovation.
- ▶ **Data:** management practices survey on 2,000 Indian firms matched to product-level sales and firm characteristics.
- ▶ **Facts:** we document key facts about management and product innovation.
- ▶ **Model:** we build a model supporting the mechanism that management matters for the conversion of ideas into products.

Preview of facts

Better-managed firms have:

1. Higher initial revenue per product
2. More R&D
3. Same creation and destruction rates
4. Higher revenue per product
5. More products

Related literature

- ▶ **Innovation, firm growth and creative destruction:** Aghion & Howitt (1992); Klette & Kortum (2004); Lentz & Mortensen (2008); Akcigit & Kerr (2018); Acemoglu et al. (2018); Garcia-Macia, Hsieh & Klenow (2019); Akcigit, Alp & Peters (2021).
- ▶ **Multi-product firms and product introduction:** Bernard, Redding & Schott (2010); Goldberg, Khandelwal, Pavcnik & Topalova (2010); Hottman, Redding & Weinstein (2016); Argente, Baslandze, Hanley & Moreira (2023); Berlingieri et al. (2025); Bloom et al. (2025).
- ▶ **Management and firm performance:** Bloom & Van Reenen (2007); Bloom, Eifert, Mahajan, McKenzie & Roberts (2013); Bloom, Sadun & Van Reenen (2016); Bruhn, Karlan & Schoar (2018); Bloom et al. (2019); Giorcelli (2019).

Data

Data

India Management and Growth Survey (IMGS):

1. Sampling frame: 15,891 manufacturing firms with financial data in Prowess
2. Sample: 2,324 firms with a management score; 89% manufacturing, median employment 270, 71% family owned, 58% exporters
3. Reference year: 2021–22 for 1,029 firms, 2022–23 for 1,296 firms
4. Respondents: managers (80%), finance staff (11%), CFOs (4%), CEOs (3%)

Survey questions: Management practices

Module based on US MOPS (Bloom et al., 2019) to capture the presence of management practices using close-ended questions

10. What best describes what happened at this business when a problem in the production process arose?

Examples: Finding a quality defect in a product or a piece of machinery breaking down.

*Please select one option**

10a. We fixed it but did not take further action	
10b. We fixed it and took action to make sure that it did not happen again	
10c. We fixed it and took action to make sure that it did not happen again, and had a continuous improvement process to anticipate problems like these in advance	
10d. No action was taken	

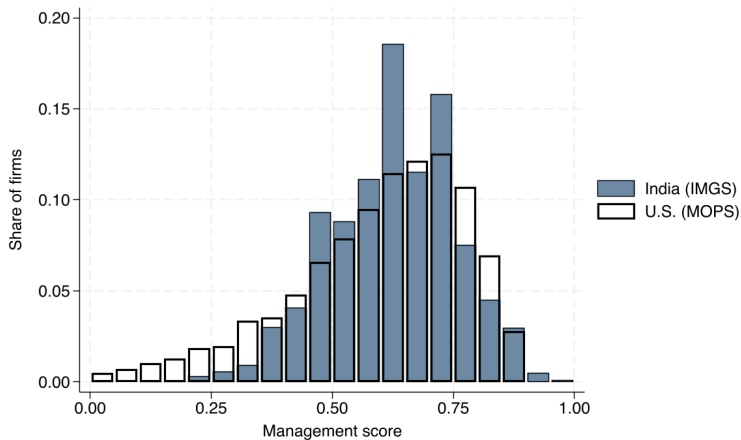
Structured management practices

- ▶ We closely follow the scoring mechanism of Bloom et al. (2019)

	Question	Score
1	In 2015, what generally best describes what happened at this business when a production problem arose?	
a	We fixed it but did not take further action	1/3
b	We fixed it and took action to make sure that it did not happen again	2/3
c	We fixed it and took action to make sure that it did not happen again, and had a continuous improvement process to anticipate problems like these in advance	1
d	No action was taken	0

- ▶ 8 questions on three dimensions of management (Monitoring, Targets & Incentives)
- ▶ complete responses: at least 5 of the 8 questions
- ▶ overall management score computed as the unweighted average of all questions, normalized to have mean zero and standard deviation one
- ▶ low scores indicating lower use of structured management practices

Distribution of the aggregate score



Note: India is IMGS firms with a management score, $N = 2,548$. The U.S. is MOPS 2015 manufacturing establishments. The y axis is the share of firms in each of 20 equal-width bins. The x axis is the management score on $[0, 1]$.

Products

Product-level sales (Prowess):

1. Source: product schedules in annual reports; sales value by product
2. Classification: product tree built on the HS system; firms report at 2 to 20 digits (most at 8 to 12). We take the code directly as reported.
3. Product: 4,406 unique codes
4. Coverage: 2,173 of the 2,324 surveyed firms report products in 2022–24; median 3 products per firm-year

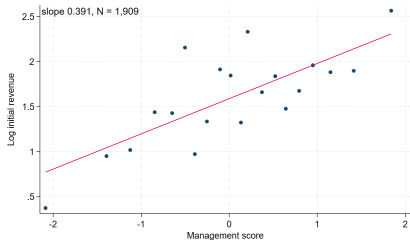
Summary statistics

	N	Mean	SD	Min	Median	Max
Management score	2,324	0.63	0.13	0.05	0.64	0.98
Family owned	2,323	0.71	0.45	0.00	1.00	1.00
Exporter	2,212	0.58	0.49	0.00	1.00	1.00
Employment	2,266	893	3,375	3	270	100,000
Products per firm	2,152	3.74	2.38	1.00	3.00	29.00
<i>R&D</i>						
positive R&D	2,220	0.14	0.34	0.00	0.00	1.00
R&D / sales (%)	301	0.90	1.41	0.00	0.37	7.85

Note: IMGS firms with a management score, 2022. Management score on $[0, 1]$. Family ownership, exporting (positive sales outside India) and employment from the survey. Products: product codes with a row in the Prowess product panel. R&D from Prowess financials. R&D/sales among firms with positive R&D, winsorised at the 99th percentile.

Facts

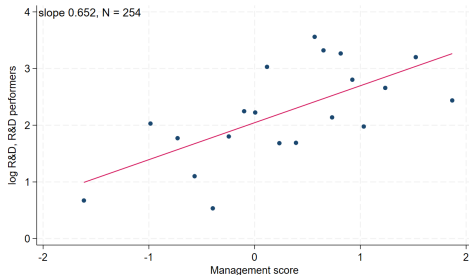
Fact 1: Better-managed firms have higher initial revenue per product



Log initial revenue	(1)	(2)	(3)	(4)
Management score	0.385*** (0.073)	0.388*** (0.073)	0.387*** (0.073)	0.385*** (0.073)
Firm age (log)		0.254** (0.117)	0.267** (0.118)	0.269** (0.118)
Family owned			-0.209 (0.155)	-0.216 (0.155)
Exporter				0.099 (0.153)
$\exp(\beta) - 1$	47.0%	47.4%	47.3%	47.0%
Observations	1,909	1,909	1,909	1,909

Note: Left figure: the y axis is log initial revenue of a product and the x axis is the standardized management score. Right table: industry, launch-year and state fixed effects, standard errors clustered by firm in parentheses.

Fact 2: Better-managed firms do more R&D



	Positive R&D	Log R&D
Management score	0.028*** (0.007)	0.571*** (0.202)
Firm age (log)	0.066*** (0.013)	1.167*** (0.250)
Family owned	-0.000 (0.017)	-0.497* (0.295)
Exporter	0.006 (0.018)	-0.091 (0.306)
$\exp(\beta) - 1$	2.8 pp	77.0%
Observations	1,984	254

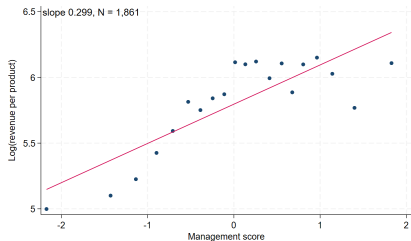
Note: Left figure: the y axis is log R&D expenditure in 2022 among firms with positive R&D and the x axis is the standardized management score. Right table: column (2) is firms with positive R&D. Industry, reference-year and state fixed effects, robust standard errors in parentheses.

Fact 3: Better-managed firms have the same creation and destruction rates

	All	Management score quartile			
		Q1	Q2	Q3	Q4
Creation rate	0.15	0.15	0.16	0.15	0.14
Destruction rate	0.16	0.17	0.17	0.16	0.15
Firms	2,197	546	560	542	549
Firm-year pairs	5,859	1,454	1,476	1,469	1,460
Mean products, $t - 1$	3.7	3.5	3.6	3.8	3.9

Note: Annual flows in 2022, 2023 and 2024 for firms with products in $t - 1$, following Berlingieri, De Ridder, Lashkari and Rigo. Creation rate is products added in t over products in $t - 1$ and destruction rate is products dropped in t over products in $t - 1$. Rates are pooled within each quartile: the sum of products added, or dropped, over the sum of products in $t - 1$ across firm-years.

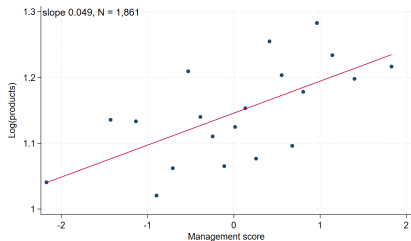
Fact 4: Better-managed firms have higher revenue per product



Log revenue/product	(1)	(2)	(3)	(4)
Management score	0.291*** (0.039)	0.290*** (0.039)	0.289*** (0.039)	0.282*** (0.039)
Firm age (log)		0.097 (0.074)	0.108 (0.074)	0.101 (0.075)
Family owned			-0.116 (0.089)	-0.118 (0.089)
Exporter				0.182** (0.079)
$\exp(\beta) - 1$	33.8%	33.6%	33.5%	32.6%
Observations	1,861	1,861	1,861	1,861

Note: Left figure: the y axis is log revenue per product in 2022, total product sales over the number of product codes, and the x axis is the standardized management score. Right table: industry and state fixed effects, robust standard errors in parentheses.

Fact 5: Better-managed firms have more products



Log number of products	(1)	(2)	(3)	(4)
Management score	0.046*** (0.014)	0.044*** (0.014)	0.044*** (0.014)	0.037*** (0.014)
Firm age (log)		0.206*** (0.023)	0.211*** (0.023)	0.204*** (0.023)
Family owned			-0.054* (0.030)	-0.055* (0.030)
Exporter				0.167*** (0.029)
$\exp(\beta) - 1$	4.7%	4.5%	4.5%	3.8%
Observations	1,861	1,861	1,861	1,861

Note: Left figure: the y axis is the log number of product codes the firm reports in 2022 and the x axis is the standardized management score. Right table: industry and state fixed effects, robust standard errors in parentheses.

Summary of facts

	Better-managed firms have	Outcome	Effect of 1 SD
1	Higher initial revenue per product	Log initial revenue	+47%***
2	More R&D	Positive R&D	+2.8 pp***
		Log R&D	+77%***
3	Same creation and destruction rates	Creation rate, Q4–Q1	–0.008
		Destruction rate, Q4–Q1	–0.015
4	Higher revenue per product	Log revenue per product	+33%***
5	More products	Log number of products	+3.8%***

Model

Households

- ▶ Time is continuous.
- ▶ A representative household supplies one unit of labor, owns all firms, and has log utility

$$U = \int_0^{\infty} e^{-\rho t} \log C_t dt.$$

- ▶ Budget constraint, with wealth A_t (the value of all firms):

$$\dot{A}_t = r_t A_t + w_t - C_t.$$

- ▶ On a balanced growth path with growth g , the interest rate is

$$r = \rho + g.$$

Final good

- ▶ A continuum of product lines $j \in [0, 1]$, all active, is combined into a CES final good ($\epsilon > 1$):

$$Y_t = \left[\int_0^1 y_{jt}^{\frac{\epsilon-1}{\epsilon}} dj \right]^{\frac{\epsilon}{\epsilon-1}}.$$

- ▶ The final good is the numeraire, so demand for line j is

$$P_t = \left[\int_0^1 p_{jt}^{1-\epsilon} dj \right]^{\frac{1}{1-\epsilon}} = 1, \quad y_{jt} = Y_t p_{jt}^{-\epsilon}.$$

- ▶ It is used for consumption and R&D: $Y_t = C_t + E_t$.

Management

- ▶ Mass M of firms.
- ▶ A firm is characterized by a time-invariant management score $z \sim \mathcal{N}(0, 1)$ and the portfolio of lines it currently leads.
- ▶ The firm chooses the **target scope** N of the portfolio it maintains and R&D **effort** e per innovation.

Innovation

- ▶ An innovation raises a line's productivity by a step λ :

$$q^+ = (1 + \lambda)q.$$

- ▶ The step depends on R&D effort and on management, which raises the return to effort:

$$\lambda(z, e) = \phi(z) e, \quad \phi(z) = \exp(\alpha + \beta z), \quad \beta > 0.$$

- ▶ Management and R&D are complements: $\lambda_e = \phi$, $\lambda_z = \beta\phi e$, $\lambda_{ez} = \beta\phi > 0$.

Innovation choice

- **Value** of an innovation, at the average target line:

$$\Omega(z, e) = \frac{Y_t (1 + \lambda)^{\epsilon-1}}{\epsilon [\rho + (\epsilon - 1)g + \tau]}.$$

- **Cost** of the program, in units of Y_t :

$$\underbrace{\Gamma(N)}_{\text{program scale}} + \underbrace{\tau N c(e)}_{\text{effort}}.$$

- The firm expands until the marginal scale cost, κX^χ , equals the net value of an innovation, $\Omega/Y_t - c(e)$.

Creative destruction

- ▶ Innovation is undirected: creative destruction hits firms uniformly across lines.
- ▶ Every product is lost at the common hazard

$$\tau = M \int X(z) f(z) dz,$$

where $X(z)$ is the rate at which a firm of type z innovates.

- ▶ To attain optimal size N , the firm innovates at rate $X(z) = \tau N$.

Model predictions

Prediction 1: Initial revenue per product

- ▶ A launch on a line at $\tilde{q}$ earns what the line earned before, times the improvement:

$$R^0 = Y_t \tilde{q}^{\epsilon-1} (1 + \lambda)^{\epsilon-1}.$$

- ▶ The firm lands on a random line and the average line earns Y_t , so expected launch revenue is

$$\mathbb{E}[R^0 \mid z] = Y_t (1 + \lambda(z))^{\epsilon-1}.$$

- ▶ Better-managed firms have a larger step λ , so they have higher initial revenue per product:

$$\frac{d \ln \mathbb{E}[R^0 \mid z]}{dz} = \frac{(\epsilon - 1)\lambda'}{1 + \lambda} > 0.$$

Prediction 2: R&D

- ▶ R&D spending is the cost of the program plus the effort cost of each innovation:

$$RD(z) = Y_t \left[\underbrace{\Gamma(N^*)}_{\text{program scale}} + \underbrace{\tau N^* c(e)}_{\text{effort on each innovation}} \right].$$

- ▶ The firm runs the program until the cost of one more innovation equals its net value, and keeps N^* products alive:

$$\kappa X^\chi = \Omega/Y_t - c(e), \quad N^* = X/\tau.$$

- ▶ Better-managed firms run more innovations (X rises) and put more effort into each (e rises).

Prediction 3: Creation and destruction rates

- ▶ A firm's products arrive at its innovation rate and each is lost at the common rate, so the count settles around the scope it chose:

$$N_f \uparrow \text{ at rate } X(z), \quad N_f \downarrow \text{ at rate } \tau N_f, \quad \mathbb{E}[N_f \mid z] = N^*(z) = X(z)/\tau.$$

- ▶ Creation and destruction rates are flows over the stock:

$$CR = \frac{X}{N^*} = \tau, \quad DR = \frac{\tau N^*}{N^*} = \tau.$$

- ▶ Both equal the economy-wide rate τ , whatever the firm's management.
Better-managed firms add more products, but in proportion to a larger portfolio.

Prediction 4: Revenue per product

- **Revenue per product:** each owned product started with higher initial revenue and fades at the same rate at every firm, so better-managed firms also sell more per product:

$$\mathbb{E}[R_f/N_f \mid z, N_f > 0] = \underbrace{Y_t(1 + \lambda(z))^{\epsilon-1}}_{\text{launch revenue}} \underbrace{\frac{\tau}{\tau + (\epsilon - 1)g}}_{\text{age erosion, same for all firms}}$$

Prediction 5: Number of products

- ▶ The net value of an innovation rises with management: a larger step makes each innovation worth more after its effort cost.
- ▶ Optimal scope depends on the innovation choice (X) and on creative destruction (τ):

$$X = \left(\frac{\overbrace{\Omega/Y_t - c(e)}^{\text{net value of an innovation}}}{\kappa} \right)^{1/\chi}, \quad N^* = \frac{X}{\tau}.$$

- ▶ **Prediction:** better-managed firms hold more products, dampened by χ .

Summary

- ▶ Better-managed firms have higher initial revenue per product.
- ▶ They do more R&D.
- ▶ They have the same creation and destruction rates.
- ▶ They have higher revenue per product and more products.
- ▶ Consistent with management working through development before launch.

Conclusion and next steps

- ▶ **Headline:** one SD of management raises launch revenue by 47%, revenue per product by 33% and R&D by 77%. Creation and destruction do not move.
- ▶ **Mechanism:** management raises the return to R&D effort. More effort goes into each idea before launch, and each idea becomes a better product.
- ▶ **Next steps:**
 - ▶ Calibration: the step and the scope-cost curvature to the five facts.
 - ▶ Competition and customers: whether better products displace rivals or deepen existing customer relationships.
 - ▶ R&D intensity: a fixed cost of running a program, to match the extensive margin and the R&D intensity gradient in the data.

Thank you

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